An Integrated World Model for Strategic Planning and Scenario Analysis

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Abstract

This paper describes an integrated simulation model of the world in which the economy, climate and the carbon cycle, warming damages, natural resources, society and politics, interstate relations, culture, and crisis and finance mechanisms are coupled within a single computational loop. The model is organized as a system of interacting countries (57 country agents plus several regional aggregates) and advances year by year: each block updates its state annually, and the results are passed to the next year and to adjacent blocks. Its aim is not to deliver a point forecast of the future but to provide a tool for strategic planning and scenario analysis: playing out coherent development scenarios, comparing alternative policy decisions, quantifying uncertainty through ensembles, and detecting weak signals of approaching structural shifts — including for the management of systemic risks. We set out the problem and the model's place among existing approaches, its architecture and the mathematics and empirical basis underlying each block, the calibration and validation procedure with the resulting metrics, sensitivity analysis and ensemble uncertainty quantification, an explicit list of limitations, and typical use cases. The economic and climate blocks are calibrated on 1990–2023 historical data and pass retrospective validation; the society, politics, and geopolitics blocks are assessed against a more modest criterion — improvement over naive benchmark forecasts.

Keywords: integrated model, strategic planning, scenario analysis, uncertainty quantification, risk management, weak signals.

1 Introduction

Strategic planning and scenario analysis over long horizons — from government policy to corporate strategy — require a tool in which climatic, economic, and socio-political processes are considered jointly rather than in isolation; the same need arises in the management of systemic risks. In practice, an analyst must rely on a patchwork of disparate models: climate models describe atmospheric physics but contain almost no finance or politics; integrated assessment models (for example, the DICE, RICE, and FUND families) couple the economy and climate but typically lack sovereign finance, interstate relations, and social dynamics [15]; macroeconomic models describe the monetary sphere in detail but ignore natural capital. As a result, the mutual influence of these domains — for example, the path from warming to falling crop yields, rising prices, social tension, and on to political instability and conflict — falls outside the scope of any single model.

Yet it is precisely the coupling of these channels that determines how long-run development scenarios unfold and where systemic risks accumulate. The absence of a coherent model spanning climate, the economy, society, and geopolitics at once is a substantive gap, felt in strategic planning and risk management alike, where the task is not to assess an isolated shock but to trace its propagation across interconnected subsystems and to compare alternative development trajectories. This paper describes a model built to fill

Table 1: Coverage comparison with representative integrated and world-system models (schematic; ~ — limited support).

Capability	DICE	GCAM	MESSAGEix	World3	GIM
Climate and carbon cycle	yes	yes	yes	no	yes
Economy and prices	yes	yes	yes	~	yes
Natural resources	no	yes	yes	yes	yes
Society and politics	no	no	no	~	yes
Geopolitics and conflict	no	no	no	no	yes
Discrete crises and finance	no	no	~	no	yes
Large-language-model agents	no	no	no	no	yes

that gap. We emphasize a principled stance from the outset: the model does not claim to be an infallible predictor. Assessments of multi-year forecasts show that even professional growth forecasts at horizons of two to five years often fail to beat a naive extrapolation of the mean [5]. A realistic goal therefore lies not in point prediction but in *the correct consistency of the model with its own measure of uncertainty* and in the ability to play out scenarios coherently.

The novelty of this work lies not in any single block but in the breadth of coupling. Integrated assessment models (DICE, FUND) and large modeling frameworks (GCAM, MESSAGEix, REMIND) couple the economy and climate in detail but contain no sovereign finance, interstate relations, or socio-political dynamics; the classic world-system models (World3) lack an explicit price economy and a modern-grade climate. The distinguishing feature of the proposed model — the Global Integrated Model (GIM) — is that it carries the loop through to society, geopolitics, and discrete crises, and embeds large-language-model agents that play as countries (Table 1, Section 7); a controlled benchmark that quantifies this advantage against sectoral models is given in Appendix B.

The coverage comparison (Table 1) is usefully complemented by a comparison along *methodological axes*: the listed models address different problems, so a direct "better–worse" ranking is inappropriate, and the distinction is better drawn by solution paradigm, treatment of uncertainty, and object of validation. In terms of *solution paradigm*, the DICE / RICE / FUND family and REMIND rest on welfare optimization (finding an optimal policy trajectory), GCAM and MESSAGEix on a detailed equilibrium of energy and land-use systems, and World3 on the system dynamics of global aggregates, whereas the proposed model is a recursive simulation of interacting countries with adaptive expectations. This implies a division of strengths: the assessment models and energy frameworks are *deeper* within their domains (optimal policy, sectoral energy detail), and the proposed model does not replace them. Its distinction appears along three dimensions at once: *(i)* breadth of coupling, carrying the single loop through to society, geopolitics, and discrete crises — quantified against sectoral models in a controlled benchmark in Appendix B; *(ii)* uncertainty as a first-class object — ensembles and history matching are built into the workflow rather than added post hoc; and *(iii)* the testability of non-standard blocks (conflict-risk ranking by AUC, an identified lagged channel between the economy and society), which comparable models simply do not contain. The model thus occupies the niche of a tool for broad scenario coverage and shock-propagation analysis, complementing rather than supplanting specialized models within their domains.

2 The model and its architecture

2.1 General design

The model represents the world as a collection of countries evolving year by year. The state of each country comprises economic, resource, social, political, and military variables, as well as bilateral relations with other countries; above the countries sit global variables of climate and the carbon cycle and world resource prices. One model step is one year: blocks update in a fixed order in which the output of one block serves as the input to the next, after which the year's results are passed forward to the following year. The computational core is deterministic; uncertainty is introduced through ensemble runs with random parameter draws from justified ranges (Section 3).

Below we describe the model's blocks, grouped thematically. For each we indicate what it computes, on what mathematics and data it rests, and what it provides to the rest of the model.

2.2 Economy

Each country's output is determined by three factors: accumulated capital, employed population, and consumed energy. The production function is specified as a nested constant-elasticity-of-substitution function (a *capital–labor–energy* form), normalized so that at the calibration base point it coincides with a Cobb–Douglas production function [1]:

$$Y = A\,\Theta\,K^{\alpha}L^{\beta}E^{\gamma}, \qquad \alpha = 0.30,\ \beta = 0.60,\ \gamma = 0.042, \tag{1}$$

where A is total factor productivity and Θ is the technology level. The nested form, with a capital–energy elasticity of substitution $\sigma_{KE} = 0.4$, captures how hard it is to substitute one factor for another: as energy becomes more expensive, the economy shifts toward capital and reduces energy use. This very mechanism allows a carbon price to restructure the economy realistically. Capital accumulates by the standard rule

$$K_{t+1} = (1-\delta)\,K_t + s\,Y_t, \qquad \delta = 0.05, \tag{2}$$

where the savings rate $s \approx 0.24$ is modulated by social stability and tension. Productivity A grows at a baseline rate of about one percent per year, responding additionally to research spending, technology diffusion through trade, and convergence toward the technological frontier; for forward projections its drift is anchored to the SSP2 socio-economic development scenario [17]. The energy, resource, and capital markets clear through prices: energy demand and investment respond to the price of energy and the cost of capital with calibrated elasticities. Inflation (a Phillips curve with energy cost pass-through), unemployment (Okun's law), the central-bank rate (a Taylor rule, [23]), the government budget and debt, and trade flows are tracked separately. Inflation additionally carries a weak quantity-theory term: excess growth of broad money (the endogenous stock of bank deposits) above a stable-velocity reference is passed through to prices, with a pass-through coefficient ($\lambda \approx 0.027$) estimated from a cross-country panel that is small by construction — consistent with the flat post-1990 Phillips curve — and whose estimation independently confirms the model's inflation-persistence anchor. The economic block reproduces the last decade of actual history and is therefore not speculative but empirically anchored.[1]

[1] Calibration sources for the economic block: the Penn World Table (factor shares and total factor productivity); the World Bank's World Development Indicators (gross product, population, energy use, military spending); the forward productivity drift is anchored to the SSP2 socio-economic development scenarios (IIASA database). A full list of variables with dataset versions and references is given in the repository file `data/external/SOURCES.md` (Section 7).

2.3 Climate and the carbon cycle

Output is accompanied by carbon-dioxide emissions, to which a land-use source is added. Emissions accumulate in the atmosphere and decay over several characteristic timescales; other greenhouse gases are accounted for, and the resulting warming is computed by a standard two-reservoir surface–ocean heat-balance model. The block is calibrated to mainstream scientific assessment: it reproduces the accepted equilibrium climate sensitivity to a doubling of carbon-dioxide concentration ($3.0\,^\circ$C) and tracks the observed temperature and carbon record from 1990 to 2023 [7, 11].[2] The strongest self-amplifying feedbacks (permafrost thaw, abrupt carbon release) are deeply uncertain and are therefore excluded from the headline run by default and explored only as tail risks. This is a fast calibrated approximation, appropriate within a model of the economy and society but not a substitute for a full Earth-system model.

2.4 Warming damages and the cost of carbon

Warming returns to the economy as damage: higher temperature reduces output. A quadratic damage function reducing the level of output is adopted,

$$\Omega(T) = 1 - 0.006\,T^2, \tag{3}$$

corresponding to losses of $2.4\,\%$, $5.4\,\%$, and $9.6\,\%$ of gross product at warming of $+2$, $+3$, and $+4\,^\circ$C. This estimate lies in the middle of the empirical range: about two and a half times higher than in DICE-2016R2 and below the central estimate of the Howard–Sterner meta-analysis [3, 10]. That this difference resides in the damage function rather than the valuation machinery is confirmed by a direct cross-model check: fed DICE's damage coefficient and discounting, the model's own marginal-pulse engine recovers DICE-2016R2's social cost of carbon (approximately $31 per ton of carbon dioxide, converging to $32–33 over a multi-century horizon), so the higher headline cost of carbon follows entirely from the higher, empirically grounded damage function. From the damage function, the social cost of carbon — the economic damage from emitting one additional ton of carbon dioxide, a standard reference point for climate policy — is computed by the marginal-pulse method. This quantity depends substantially on the discounting scheme and on the structure of future growth; results are reported in Section 3.[3]

2.5 Resources

For each country, stocks, production, and consumption of energy, food, and metals are tracked; world prices rise when demand exceeds supply, and this price is passed into inflation and on into the economy. The resource market clears by price with a calibrated demand elasticity; a slow-adjustment rule is provided as a fallback. Resources are represented in aggregate rather than as detailed sectoral energy systems.[4]

2.6 Society, politics, geopolitics, and culture

For each country the model tracks trust in government, social tension, income inequality, institutional legitimacy, and protest pressure. These variables respond to the economy

[2] Sources for the climate block: the equilibrium-sensitivity estimate — the IPCC Sixth Assessment Report (Working Group I); the global emissions series — the Global Carbon Project (Global Carbon Budget) and the Our World in Data compilation; the surface-temperature series for 1990–2023 — NOAA/HadCRUT5.

[3] Calibration of the damage function draws on the Howard–Sterner meta-analysis and the panel estimates of Burke and coauthors; the social-cost-of-carbon reference points are the estimates of the U.S. Environmental Protection Agency (EPA, 2023) and the RFF-SP project.

[4] The resource block is specified in aggregate: energy indicators draw on the World Development Indicators; price elasticities are set from generalized market estimates rather than a dedicated sectoral dataset.

— rising inflation and unemployment raise tension and erode trust, and high inequality amplifies this effect — and, in turn, affect stability and crisis risk. Each pair of countries is linked by a relation: trade volume, the presence of an alliance, the level of conflict or tension, a state of war, and sanctions. Each country's military capability is anchored to observed measures of national power rather than to arbitrarily assigned values [20]. A small set of cultural traits (chiefly individualism, uncertainty avoidance, long-term orientation, and hierarchy), together with the character of the political regime, influence economic and social behavior [9]. Unlike the economy and climate, no single authoritative dataset exists for these links, so they are built on expert judgment and the general evidence base and are assessed against a more modest criterion (Section 3).[5] Traits that did not in fact affect outcomes have been removed from the model, so that it does not mimic cultural detail it does not use.

2.7 Risk, crises, and finance

On top of the smooth annual dynamics, the model recognizes and applies discrete crises — debt, currency, political-regime collapse, and extreme climate events — each triggered when the corresponding threshold is crossed. Crisis severity may be drawn from a heavy-tailed distribution (most events are moderate, rare ones catastrophic), matching the observed distribution of real crises and wars [22, 25]; this mode is optional and is explored as uncertainty. The financial side is kept with strict accounting: every change in debt is balanced by a counter-flow. Government finances are closed, and the private sector is described by a full bank balance sheet in which every loan creates a matching deposit, so that the money stock is identically equal to deposits and loans; excessive borrowing raises the cost of credit and restrains the economy — a financial accelerator [2, 8].[6]

2.8 Block interactions

Connectivity is the model's defining property. The economy generates emissions; emissions determine warming; warming, through the damage function (3), reduces output; the carbon price raises the cost of energy and, through substitution in (1), lowers emissions. Economic hardship raises social tension and lowers trust, which changes the savings rate in (2) and raises crisis risk; a crisis hits output and capital and is transmitted to trading partners. Resource scarcity raises prices and inflation; conflicts and sanctions disrupt trade. A shock in one subsystem thus propagates through the others — and this coupling is exactly what gives the model its substantive value for risk analysis.

3 Calibration and validation

3.1 Data sources

Each block is anchored to external data (Table 2). The economic and climate blocks are calibrated and validated most rigorously; the socio-political and geopolitical variables are anchored to recognized indices but assessed against a softer criterion.

[5]The socio-political and geopolitical variables are anchored to recognized indices: income inequality — the SWIID database and the World Bank Gini index; legitimacy, governance quality, and political stability — the Worldwide Governance Indicators; armed conflicts — the UCDP/PRIO Armed Conflict Dataset (v. 24.1); military capability — National Material Capabilities (Correlates of War, the CINC index) and the SIPRI military-expenditure database; sanctions — the Global Sanctions Database (v. 3); cultural traits — the Hofstede indices. The mapping of each variable to its source is described in `data/external/SOURCES.md`.

[6]There is no dedicated calibration dataset for the risk-and-finance block: the heavy-tailed crisis-severity distributions and the stock-flow-consistent accounting rest on the methodological literature cited in the text rather than on a single external source.

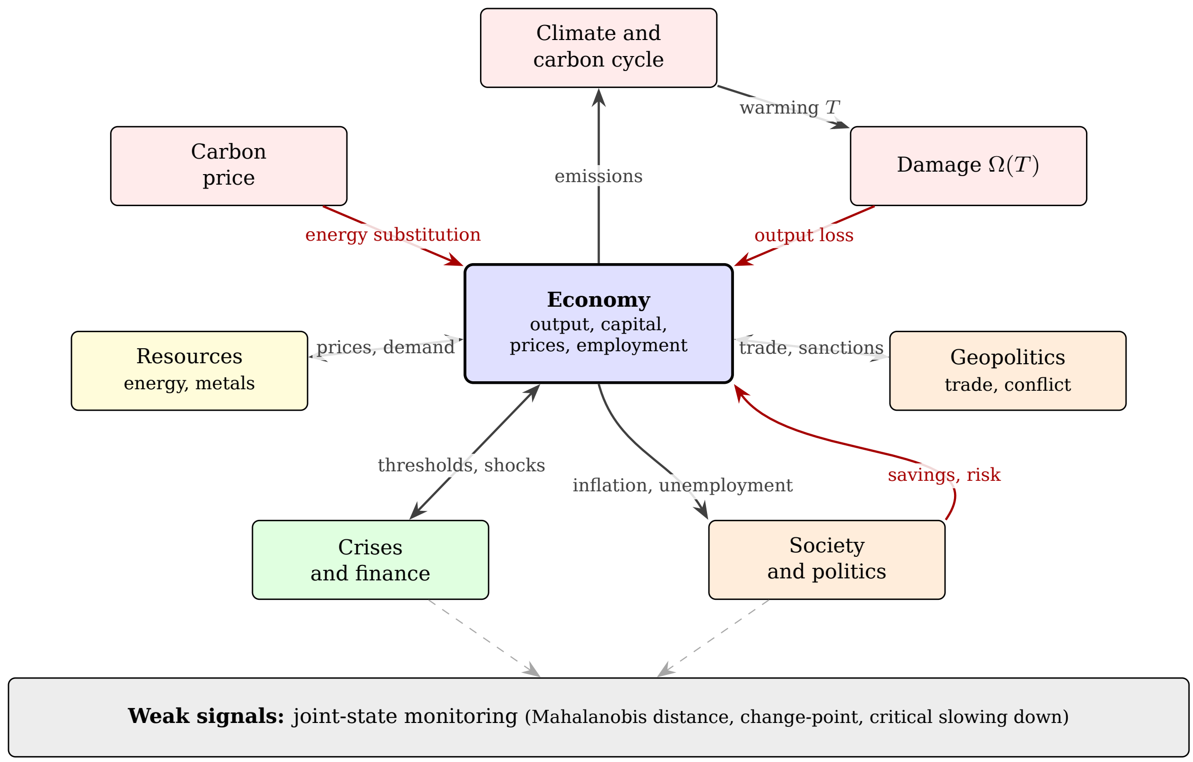

Figure 1: Model architecture and the links between blocks. The economy is the central hub; black arrows show direct effects, red arrows show feedbacks that reduce output or savings. The closed climate loop (from emissions to warming, then to damage and reduced output) is complemented by the carbon-price channel, the resource, socio-political, and geopolitical blocks, and the discrete-crisis block. The bottom layer implements continuous monitoring of the joint state to detect weak signals.

3.2 Calibration principles and integrity control

Calibration follows several rules. First, every optional mechanism is implemented as a switch, off by default unless anchored by historical evidence; thus the model's headline ("base") behavior remains tied to data, while speculative mechanisms are explored separately as uncertainty. Second, parameter estimation uses not point fitting but *history matching*: instead of seeking a single "best" parameter set, regions of parameter space incompatible with observations are discarded, leaving a posterior region of "not-yet-ruled-out" values — a standard approach for computationally expensive simulation models [26]. Third, the climate coefficients derived from the initial state are tightly bound to a checksum of the input data and cannot be changed by hand; the integrity of this link is checked automatically. Finally, regression control is in force: the significant retrospective-validation quantities are pinned, and any change to the core must either preserve them within tolerance or be accompanied by an explicit recalibration.

3.3 Resulting metrics

A validation summary across all blocks is given in Table 3, with the key results discussed below. Retrospective validation over 2015–2023 yields root-mean-square errors for gross product, world emissions, and temperature. The move from the former Cobb–Douglas production function to the closed core with a nested substitution function and market clearing improved the fit to history (Fig. 2a): the product error fell from 1.03 to 0.59 trillion USD, and the emissions error from 1.61 to 1.15 gigatons.

Comparison with benchmark forecasts reveals the structure of this accuracy. The

Table 2: Empirical basis of the model's blocks.

Block	Data source	Purpose
Economy	Penn World Table, World Development Indicators	factor shares, growth, debt
Climate and carbon	IPCC AR6 assessment, Global Carbon Budget	sensitivity, T and CO_2 paths
Cost of carbon	EPA-2023, RFF-SP	reference for cross-checking
Forward growth	SSP2 scenarios	forward productivity drift
Conflict	UCDP Georeferenced Event Dataset	conflict-risk validation
Inequality	SWIID	income inequality
Institutions	Worldwide Governance Indicators	legitimacy, governance quality
Military power	composite national-power index	country capability
Culture	Hofstede indices	cultural traits

Table 3: Validation summary by model block: data source, validation criterion, and the achieved result. The economic and climate blocks are checked by strict retrospective validation; for the others the criterion is softer — cross-checking against published estimates, improvement over a naive benchmark, or channel identification. The comparison of the current core with the former one (Cobb–Douglas) is shown in Fig. 2a.

Block	Source	Criterion	Result
Economy	PWT, WDI	retrospective RMSE, 2015–2023	GDP 0.59 trillion USD (Cobb–Douglas 1.03)
Climate and carbon	IPCC AR6, GCB	retrospective RMSE, 2015–2023	CO_2 1.15 Gt; T $0.135\,°C$
Cost of carbon	EPA-2023, RFF-SP	published cross-check; DICE engine reproduction	$\approx\$140$/t CO_2 (range 140–380); recovers DICE-2016R2 \$32–33 under DICE inputs
Conflict	UCDP/PRIO	ranking vs. registry	AUC 0.736 $[0.59;\ 0.86]$; BSS $+0.143$; $p{\approx}0.001$
Social block	SWIID, WGI	channel identification (Sec. 4)	significant lagged channel; regression $p < 10^{-12}$

model *beats* the persistence forecast for temperature (a skill score of about $+0.14$) and marginally for product, but *loses* to a naive linear extrapolation for product and emissions over this short, smooth interval. This is to be expected and is consistent with assessments of professional forecasts [5]: a structural model raises skill where dynamics matter (climate) and loses to trivial extrapolation where a short-run smooth aggregate is hard to beat.

A different criterion is applied to the conflict block. The countries' conflict-proneness score is compared with the historical registry of armed conflicts over 1990–2023 across 57 countries [21]. The framing of the test is essential: what is evaluated is not an output fitted to these same data but an *input* risk indicator computed from observed country characteristics and not tuned to the conflict registry. Hence there are no parameters trained on the target variable and no associated information leakage, and a rolling train/test split is unnecessary; the full set of 57 countries is evaluated as a whole. The model confidently ranks conflict-affected countries above calm ones: the area under the characteristic curve is 0.736 (bootstrap $95\,\%$ interval $[0.59;\ 0.86]$, $20\,000$ pair resamples with replacement [6]), and a permutation significance test (random label shuffling, $50\,000$ repetitions) gives $p \approx 0.001$, so the ranking is decisively better than chance. The Brier skill score relative to the base-rate forecast is $+0.143$ (a positive value indicates predictive value). This is a reasonable bar for a block of this kind: the model is suitable for assessing relative risk but not for

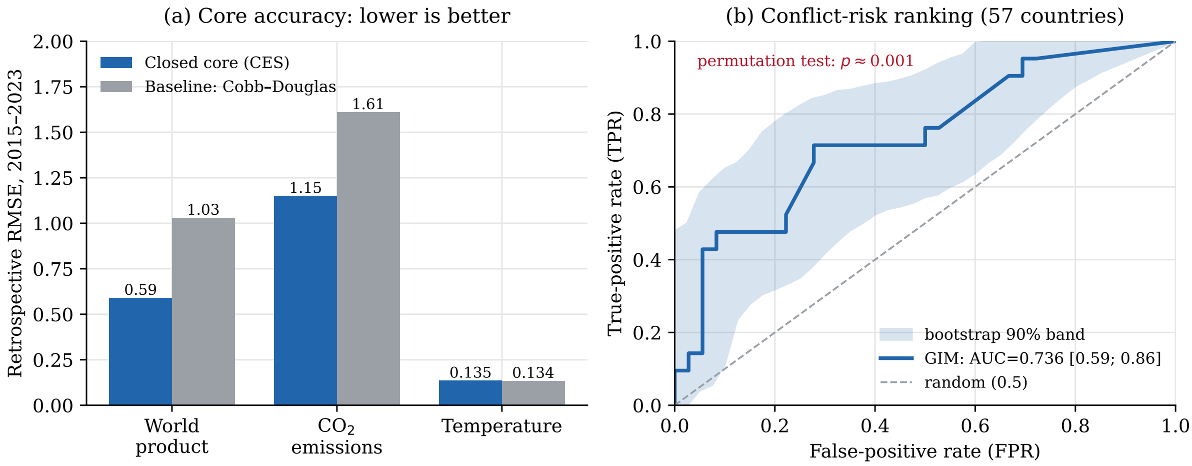

Figure 2: Retrospective-validation results. *(a)* Root-mean-square error of the 2015–2023 retrospective check for the closed core (nested CES) compared with the former Cobb–Douglas core, for world product, CO_2 emissions, and temperature (lower is better). *(b)* ROC curve of the conflict block: ranking 57 countries by conflict-proneness against the registry of armed conflicts of 1990–2023 yields an area under the curve of 0.736 (the bootstrap $90\,\%$ band is shaded; $95\,\%$ CI $[0.59; 0.86]$; permutation test $p \approx 0.001$) versus 0.5 for random guessing.

predicting specific wars on specific dates.

The social cost of carbon is sensitive to the discounting scheme. Under a modern Ramsey scheme with a near-zero pure rate of time preference of about two percent it is roughly \$140 per ton — within the broad modern range, slightly below the central EPA-2023 and RFF-SP estimates (about \$190) [16, 24]; under Nordhaus-style discounting it is about \$43. Importantly, this quantity is *genuinely* sensitive to the structure of future growth: across different calibrations of the economic core it varied over a range of roughly \$140–380. We report this value together with its sensitivity rather than tuning it to a target; that sensitivity is itself a standalone result.

4 Uncertainty, sensitivity, and parameter identification

A key question is whether the model's many parameters are identified by data rather than set by hand. Each key parameter is equipped with a prior distribution — a central value, a plausibility range, and a literature source (Appendix A); estimation proceeds not by point fitting but by history matching (Section 3), discarding regions of parameter space incompatible with observations. Thus the capital–energy elasticity of substitution $\sigma_{KE} = 0.4$ rests on empirical estimates in the 0.3–0.5 range, the depreciation rate $\delta = 0.05$ on Penn World Table data (3–8 %), and the damage coefficient 0.006 lies in the middle of the empirical range 0.0015–0.025, between the DICE-2016R2 estimate and the central Howard–Sterner estimate. The irreducible uncertainty of these quantities is not hidden but carried into the computation through ensembles.

Sensitivity analysis. To establish which parameters actually determine outcomes, a global screening by the Morris elementary-effects method [4, 14] was carried out over 26 key parameters for four output quantities (Fig. 3; 8 trajectories, 10-year horizon). The picture is consistent with the model's structure and with physics: temperature is determined almost exclusively by climate parameters — equilibrium sensitivity, ocean heat exchange, and surface heat capacity; world product by capital depreciation, the damage coefficient, and the energy intensity of output; world emissions by the emissions scale and the rate of structural decarbonization. These conclusions are robust to the screening budget: increasing the number of trajectories from 8 to 32, the Spearman rank correlation of the full μ^* vector with the reference run is $\rho \geq 0.99$ for all four quantities, and the leading parameters reappear in the top five across all tested random-number seeds (a complete match for temperature and emissions; for product the top four are stable, while the fifth position is exchanged between two economic parameters). Social tension, by contrast, is almost insensitive to these physico-economic parameters: its elementary effects lie at the screening noise floor (the absolute μ^* is about three orders of magnitude smaller than for the physical quantities), and their ranking — unlike the stable ranking of the physical quantities — is itself unstable to the choice of seed. Importantly, however, this does *not* mean that society is decoupled from the economy: this screening varies only the physico-economic priors, whereas tension is governed by its own (social) parameters and responds to economic conditions with a multi-year lag. We isolate this channel from economic conditions to social tension separately below (Fig. 5). The screening thus confirms that the significant quantities are governed by substantively appropriate and empirically anchored parameters, and this conclusion is not an artifact of the limited computational budget.

Ensemble uncertainty quantification. Because the core is deterministic, uncertainty is introduced by running a Monte Carlo ensemble (500 members) with random parameter draws from their prior ranges. The result is not a single trajectory but a distribution of outcomes (Fig. 4). Throughout, "world product" denotes the *aggregate gross product of the* 57 *modeled countries* (the world's largest economies, accounting for the bulk of global output), not the gross product of all economies on the planet; in the 2026 base year it is about 109 trillion USD. At the ten-year horizon its median is about 116 trillion USD, with an interquartile range of roughly 114–118 and a 5–95th-percentile interval of roughly 110–121; the 5–95th-percentile interval of the temperature anomaly is roughly 1.0–$2.0\,^{\circ}$C and reflects chiefly the spread in equilibrium sensitivity. The reported percentiles have converged in ensemble size: as the number of members grows from 80 to 500, the median and upper percentiles of product and temperature are stable, and the Monte Carlo error of each percentile (a bootstrap over ensemble members) does not exceed ± 0.3 trillion USD for product and $\pm 0.03\,^{\circ}$C for temperature; the only quantity noticeably dependent on sample size is the lower temperature tail, which tightens from ≈ 0.9 to $\approx 1.0\,^{\circ}$C. It is precisely such intervals, rather than point values, that constitute the model's substantive result.

The channel from the economy to social tension. The low sensitivity of tension to the physico-economic *priors* (above) does not mean that society is decoupled from the economy: tension is an accumulating variable (its increment is determined by unemployment, inflation, and inequality), so its link to the economy manifests not instantaneously but with a lag, and is invisible to a fixed-horizon screening. We isolate this channel through three complementary checks (Fig. 5; the reproducible computation is described in Section 7). First, *localization*: in an ensemble that additionally varies the priors of the social block itself, final tension is significantly governed by them — by the sensitivity to inequality (Spearman rank correlation $\rho \approx +0.43$, $p < 10^{-6}$) and the strength of the trust anchor ($\rho \approx -0.50$, $p < 10^{-8}$) — whereas the physico-economic priors remain weak; that is, the screening above simply did not vary the block's direct drivers. Second, *accumulation*: the prior-induced spread of tension grows with the horizon (the across-member standard deviation increases by roughly half from 5 to 25 years), which is the signature of a lag. Third, *impulse response*: an exogenous stagflation shock (a three-year rise in unemployment and inflation) raises tension, with the response building over the first ≈ 4 years (positive for 93–$100\,\%$ of the prior-ensemble runs) and persisting for more than a decade; at longer horizons the spread grows because in some runs the shock triggers discrete crises. A distributed-lag regression of the change in tension on economic stress (within-member fixed effects, clustered errors) confirms significant transmission at lags of 0–1 years ($p < 10^{-12}$). The channel from the economy to tension is thus statistically significant

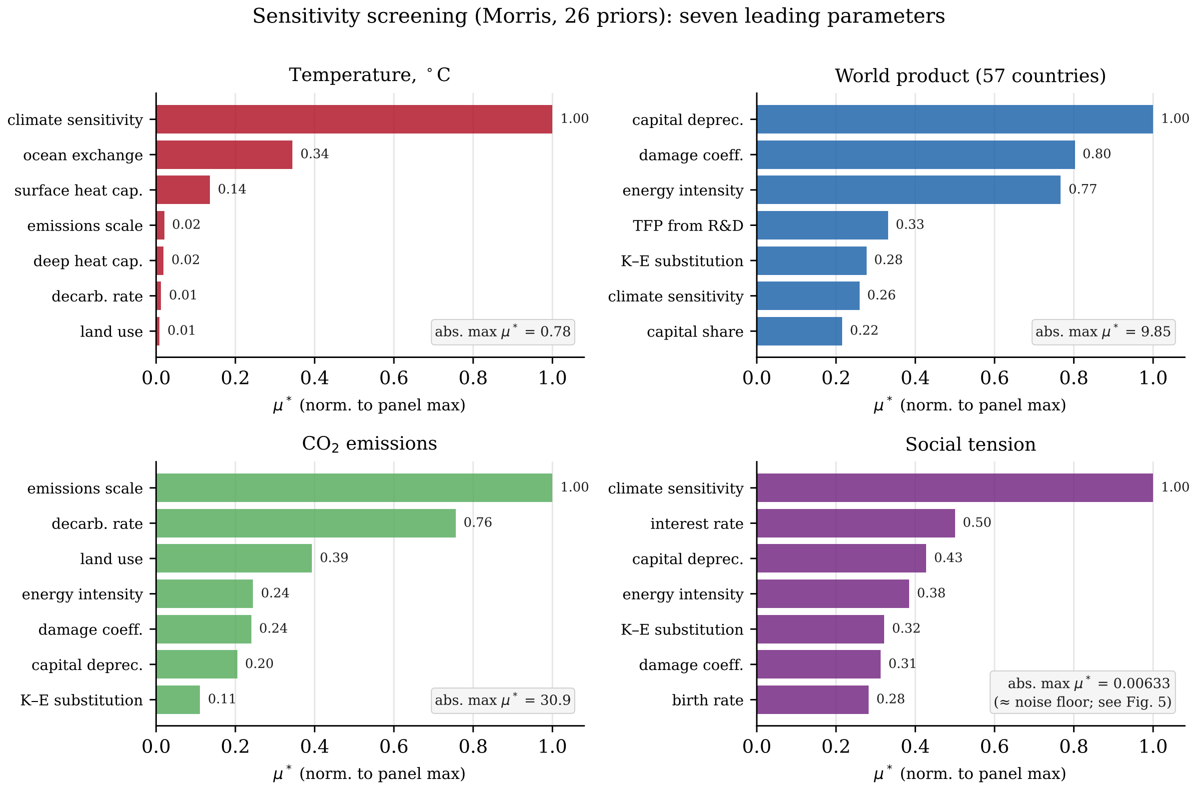

Figure 3: Global sensitivity screening (Morris method) over 26 key parameters for four output quantities (57 countries); the seven most influential parameters are shown, with μ^* normalized to the maximum within each panel and the absolute maximum μ^* indicated in the box. Temperature is governed by climate parameters, product and emissions by economic and emission parameters. For social tension the absolute μ^* is about three orders of magnitude lower (at the noise floor): it is almost insensitive to the physico-economic priors, and its own channel is examined in Fig. 5.

but mediated and lagged; it is precisely this lag, not an absence of any link, that explains the null result of the short-horizon screening.

5 Limitations

An honest list of what the model *cannot* yet do is no less important than a list of its capabilities.

Expectations are adaptive. Households and firms look to the recent past rather than to a rationally anticipated future. The model therefore does not reproduce anticipatory effects (for example, a response to a pre-announced policy before it takes effect).

Long-run growth is weakly identified. The long-run rate of productivity growth is only weakly pinned down by the available short history and, for forward projections, is anchored to an external scenario rather than derived endogenously.

The climate is described by a fast approximation. The climate block reproduces the standard indicators but is not a full Earth-system model; the strongest carbon feedbacks are deeply uncertain and are relegated to tail scenarios.

Damage estimates are uncertain across the discipline. Estimates of climate damage are uncertain across the entire field; the model carries this uncertainty as an interval rather than as a single number.

The socio-political links rest on structural assumptions. There is no single calibration dataset for the transition from conditions to social discontent, so the forms and magnitudes of the social-block coefficients are set by expert judgment. The channel from

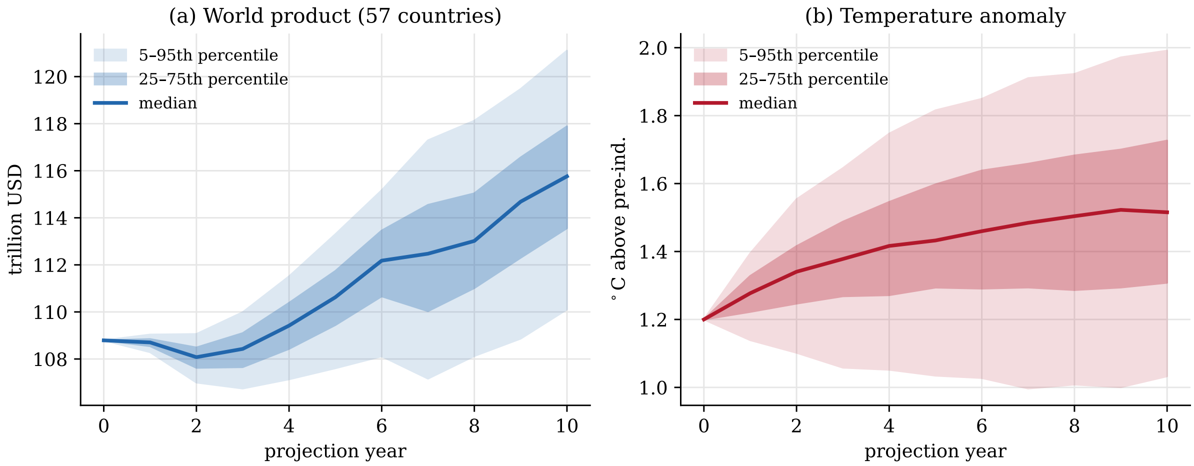

Figure 4: Ensemble uncertainty quantification (500 members): the distribution of outcomes from a Monte Carlo run with random parameter draws from the prior ranges. The median and the 25–75th- and 5–95th-percentile bands are shown for *(a)* the aggregate product of the 57 modeled countries and *(b)* the temperature anomaly.

the economy to tension is nonetheless not arbitrary: it is statistically significant, directionally correct, and lagged (Section 4, Fig. 5); what remains unestablished is the precise functional forms and lags, not the existence of the link. Conflict is intrinsically hard to predict, and the geopolitical block is suitable only for relative risk.

Crises are threshold-based. Discrete crises are specified by threshold rules and capture the correct qualitative behavior but are not derived from a single fitted crisis dataset.

The money-to-prices transmission needs further work. The government and private-sector balance sheets are closed and accounting-consistent, but a more explicit transmission from the money stock to price pressure remains a subject for further calibration.

The overall stance is unchanged: the realistic goal is to be calibrated against one's own uncertainty rather than to serve as an instrument of infallible foresight. A realistic accuracy bar for multi-year forecasts is improvement over a naive benchmark.

6 Typical use cases

The model is intended for three interrelated modes of use, in which its connectivity and proper uncertainty quantification pay off most.

Scenario analysis and strategic planning. The principal mode is comparison of a baseline trajectory with a perturbation or policy scenario: a shock or measure is introduced (for example, a carbon price, resource scarcity, or an escalation of relations between countries), and its propagation through the economy, climate, society, and geopolitics is traced. This makes it possible to compare alternative strategies and assess the attendant risks. What matter here are *relative* quantities — differences between scenario and baseline — rather than absolute point values at a distant horizon.

Ensemble computations. Because the computational core is deterministic, uncertainty is introduced by running large ensembles with random parameter draws from justified ranges. This yields not a single trajectory but a distribution of outcomes, against which sensitivity to each parameter is measured and probabilistic estimates are constructed. Scenarios are best compared as ensembles rather than as single runs.

Weak-signal detection. The built-in crisis logic is threshold-based: a crisis fires when

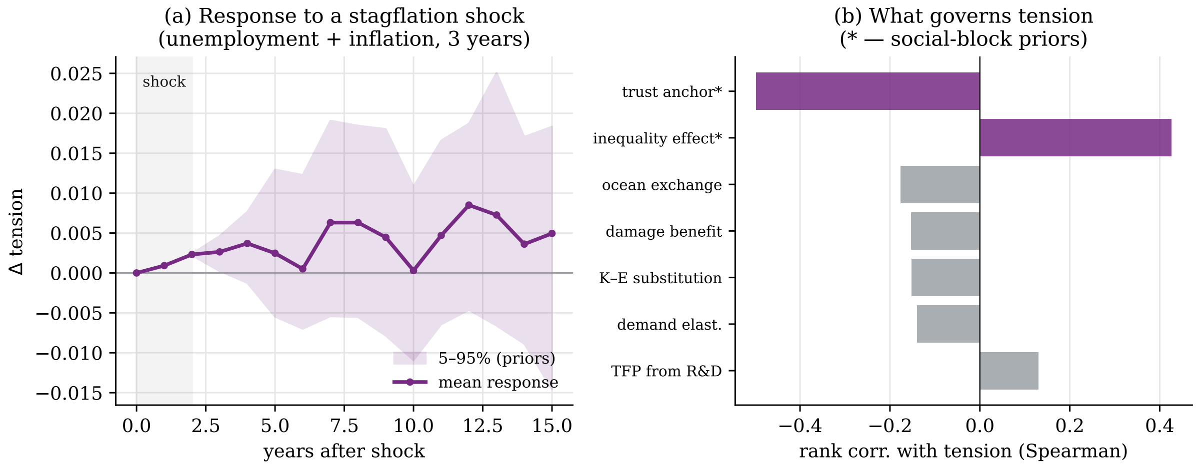

Figure 5: The influence of economic conditions on social tension. *(a)* Impulse response of mean tension to an exogenous stagflation shock (unemployment and inflation, three years): the mean curve and the 5–95th-percentile band over priors; the response builds over ≈ 4 years and persists. *(b)* Localization: the rank correlation of priors with final tension; the parameters of the social block (marked $*$) dominate, not the physico-economic ones.

a boundary is crossed. Weak-signal detection complements it: the identification of a subtle, multivariate accumulation of tension *preceding* a structural shift, before any threshold is crossed. Three complementary tools are used. The first is the Mahalanobis distance: at each point in time it is assessed how far the *joint* state (product, temperature, debt, tension, and so on, together) lies from its recent baseline distribution, accounting for correlations between dimensions; exceeding a χ^2 threshold indicates the system's transition into an unusual region of state space [13, 27]. The second is structural-break detection: a single series is searched for a regime change by change-point detection with an information-criterion penalty, yielding a probability of a shift and its most likely location [19]. The third is critical-slowing-down indicators (rising autocorrelation and variance) that grow as a bifurcation point is approached [18]. So that these tools detect deviations from the *dynamics* rather than secondarily detecting a smooth trend, the series are first reduced to stationary increments. On synthetic data and on the model's own trajectories the tools behave as expected: an injected $15\,\%$ drop in product is detected exactly at the right step, whereas a calm run stays near the theoretical false-alarm rate. The working device here, as in scenario analysis, is *relative*: comparing a scenario's weak-signal profile with the baseline's (earlier and more frequent deviations, a higher shift probability) indicates that the scenario is moving the system toward a structural transition.

7 Software implementation and the scenario environment

The model is implemented as an open codebase and is available in the repository `https://github.com/Theclimateguy/GIM`. The computational core is deterministic and reproducible: each run is written to a separate dated folder with a manifest of parameters and input-data versions, and a suite of unit tests and a validation protocol pin the key retrospective-validation quantities (Section 3). Calibration datasets are attached as immutable, checksum-bound inputs, which precludes silent divergence of results between runs. The baseline world run, scenario questions, multi-agent games, and calibration are launched through a single command-line interface.

Large language models as countries. Beyond the built-in behavioral rules, an individual country's policy may be set by a large-language-model agent. The agent receives the observed state of its country and its environment and returns a coherent strategy — domestic, foreign, financial, military, and trade policy — that is compiled into the model's formalized actions. Several background-policy modes are supported: rule, growth target, language model, and a precompiled doctrine; the user sets the depth of strategy updating (per event, periodically, or once). A methodological distinction is important here: the generative model produces only the actors' intentions and strategy, whereas their consequences are computed by the deterministic, empirically anchored core. The language model is responsible for the plausibility of behavior, the simulation core for the plausibility of the mechanics and numerical outcomes; this separation preserves reproducibility and shields the quantitative results from arbitrary judgments of the language model.

Hybrid games and "what-if" hypotheses. The environment allows joint play by humans and agents: some countries are led by users, some by language agents or built-in rules, and intentions are stated in natural language and translated into model actions. Two move modes are provided. In *action* mode a move is committed and carried into the subsequent trajectory; in *"what-if"* mode a hypothesis is played out in isolation, without irreversible consequences for the main line, which is convenient for testing counterfactual assumptions (for example, "how would the situation change if country X tightened sanctions against country Y"). Each move unfolds not into a single trajectory but into an ensemble, so that the response is measured together with its uncertainty — as in the scenario analysis of Section 6.

Interactive analytical environment. A separate command launches a local analytical dashboard that gathers scenario questions, multi-agent games, and country metrics; the games tab uses the hybrid mechanism described above. The results of each run are exported in a readable form — a summary dashboard, an analytical brief for decision-makers, and logs of policies and baseline trajectories — which makes a multi-agent episode traceable and amenable to subsequent review.

The scenario environment thus operationalizes the model's stated purpose. The coupled core turns from a computational mechanism into a platform for exercising multi-agent interaction: a collective of language agents or a mixed human–agent composition plays out specific scenarios and counterfactual hypotheses, while the user observes the propagation of shocks through the economy, climate, society, and geopolitics and the early weak signals of approaching structural shifts — within a single, reproducible, auditable loop. The role of generative AI here is auxiliary but substantive: it expands the space of plausible actor strategies without supplanting the empirically anchored mechanics of consequences.

8 Conclusion

The model presented here links the economy, climate, natural capital, society, politics, geopolitics, and culture within a single loop and thereby fills a substantive gap between specialized single-domain models. Its value for strategic planning, scenario analysis, and risk management lies not in point forecasting but in the ability to play out scenarios coherently, compare alternative decisions, express uncertainty correctly through ensembles, and detect weak signals of approaching structural shifts. The economic and climate blocks are empirically anchored and pass retrospective validation; the socio-political and geopolitical blocks, which constitute the model's distinguishing feature, are validated against a more modest but substantive criterion — improvement over naive benchmarks. Further development concerns the introduction of forward-looking (rational) expectations, a deeper grounding of long-run growth, and an explicit money-to-prices transmission; these directions are framed as limitations and subjects for future work rather than as obstacles to applying the model in the modes outlined.

A Key parameters and their identification

Each parameter is specified by a prior distribution; below are the central value, the plausibility range (roughly the 5–95th percentile of the prior), and the main source. The full set of 26 key and 294 total parameters with distributions and justifications is available in the repository (Section 7).

Parameter	Central	Range	Source
Equilibrium climate sensitivity, °C	3.0	1.5–6.0	IPCC AR6 WG1 (2021); Otto et al. 2013
Surface heat capacity	8.0	6.0–18.0	Geoffroy et al. 2013 (two-layer EBM)
Ocean heat exchange	1.0	0.4–1.3	Geoffroy et al. 2013
Emissions scale	0.98	0.90–1.05	Global Carbon Project 2023
Structural decarbonization rate	0.052	0.040–0.066	Calibration to 2015–2023 backtest
Capital share α	0.30	0.22–0.40	Penn World Table 10; Gollin 2002
Labor share β	0.60	0.50–0.72	Penn World Table 10; Gollin 2002
Energy elasticity γ	0.042	0.015–0.080	2015–2023 backtest; energy literature
Capital–energy substitution σ_{KE}	0.40	0.20–0.60	van der Werf 2008; Koetse et al. 2008
Depreciation rate δ	0.05	0.03–0.08	Penn World Table 8/9
TFP sensitivity to R&D	0.30	0.10–0.60	Calibration to backtest
Damage coefficient	0.006	0.0015–0.025	DICE-2016R2; Howard & Sterner 2017; Burke et al. 2015
Pure rate of time preference ρ	0.015	0.001–0.030	DICE-2016R2; Stern 2007
Elasticity of marginal utility η	1.45	1.0–2.0	DICE-2016R2; Stern 2007
Market demand elasticity	0.40	0.20–0.70	Labandeira et al. 2017 (meta-analysis)
Social block (structural assumptions; validated by sign and channel, Section 4)			
Tension stress from unemployment	0.010	0.007–0.014	Structural assumption (higher unemployment raises discontent)
Tension stress from inflation	0.005	0.003–0.007	Structural assumption
Inequality effect on tension	0.0005	0.0003–0.0007	Structural assumption
Trust-anchor strength	0.060	0.039–0.081	Structural assumption (reversion to equilibrium)

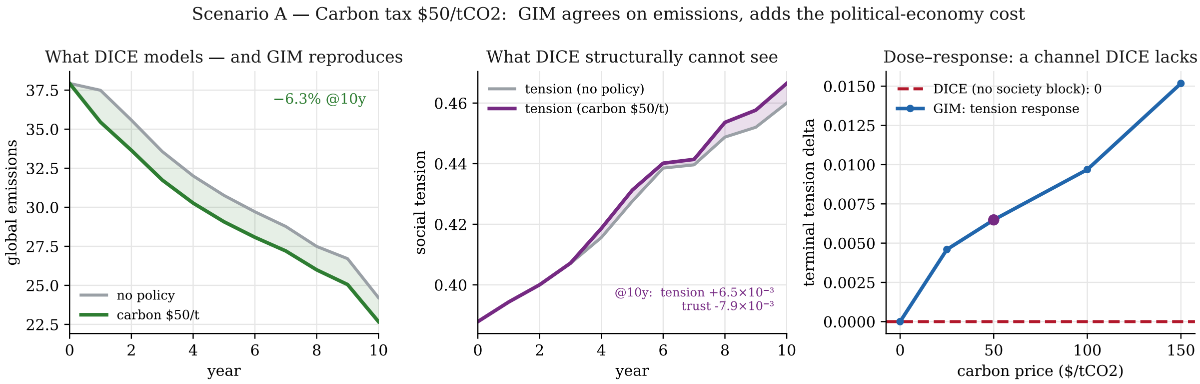

Figure 6: Carbon-tax scenario. Left: GIM reproduces the emissions decline that DICE models. Middle: GIM adds the social-tension response DICE cannot represent. Right: the response of tension to the tax rate has a non-zero GIM slope against DICE's zero response.

B Integration benchmark: where sectoral models stop

The breadth-of-coupling claim (Table 1) is made *computational* here. The design is literature-anchored rather than an internal ablation: for each of three shocks we take a published conclusion from a named sectoral model, feed GIM the *same* exogenous shock, and ask whether GIM *(a)* reproduces the sectoral first-order effect and *(b)* surfaces a cross-sector consequence the sectoral model cannot represent by construction. The unit of comparison is the *decision-relevant conclusion*, not a headline number: GIM is not claimed to compute emissions or welfare "better" than DICE, but to see a consequence DICE is structurally blind to. Runs are deterministic (the full 57-country world, a ten-year horizon, extreme events off, so the only difference between the baseline and shock arms is the injected shock); each scenario also reports a dose–response sweep on which the sectoral model's slope is, by construction, zero. The harness and figures are reproducible from the repository (Section 7).

Two methodological points are stated plainly, as they are part of the result. First, GIM's aggregate resource-*price* subsystem saturates within one simulated year, so a purely price-mediated shock does not propagate forward; each shock is therefore injected at its first cross-sector transmission variable — precisely the channel a sectoral model lacks (a consumer-price wedge for the carbon tax, an import-bill and reserve burden for the oil shock, a food-supply gap for the crop shock). Second, the three spillovers differ in magnitude and shape, and are reported as such rather than uniformly inflated.

B.1 Carbon tax (anchor: DICE)

DICE-2016R [15] finds a welfare-optimal carbon price (about $31/tCO$_2$ in 2015, rising) whose only first-order effect is lower emissions; it contains no labour market and no political economy. Fed the same $50/tCO$_2$ tax, GIM reproduces the emissions decline (about 6% over ten years) and, in addition, raises social tension and lowers government trust as the tax's pass-through to energy prices feeds inflation and unemployment — the political-economy channel behind the reversals of the French carbon and fuel tax (2018) and the Australian carbon price (2014). As the rate increases, the tension response rises monotonically (Fig. 6), while DICE's response on that axis is zero.

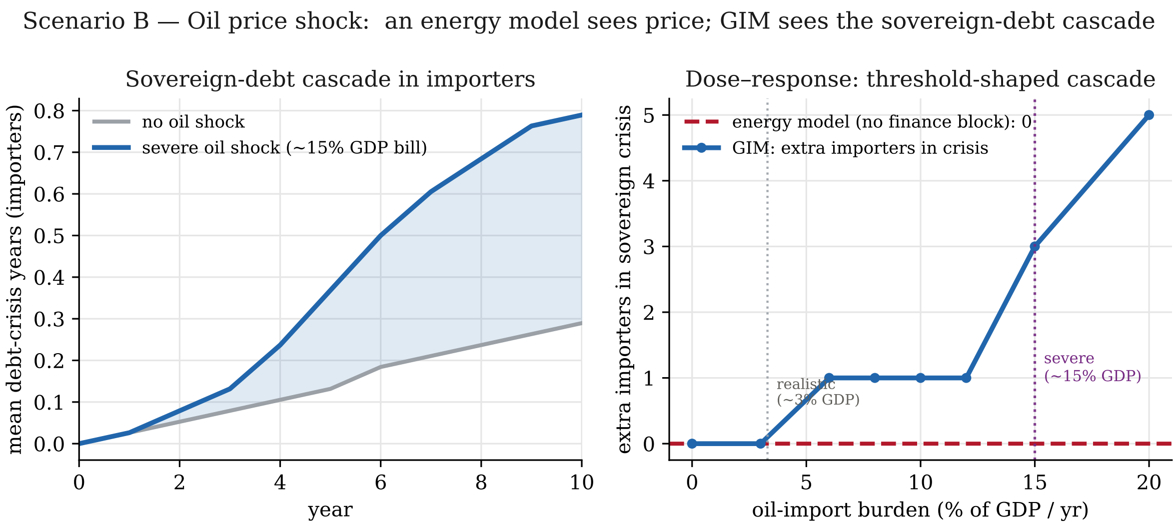

Figure 7: Oil-price-shock scenario. Left: importer sovereign-debt-crisis duration under a severe shock. Right: the response to the import burden is threshold-stepped, marking the marginal (realistic) and pronounced (severe) points against the energy-system model's zero response.

B.2 Oil price shock (anchor: energy-system models)

An energy-system model such as MESSAGEix resolves an oil-supply shock as a price rise and demand adjustment, bounded within the energy sector. GIM carries the same shock further, into sovereign finance: the higher import bill, financed by drawing reserves and issuing debt, brings the most fiscally fragile importers to their debt-crisis thresholds, and which countries tip is decided by the model's own debt and interest-rate triggers, not by hand. At a realistic shock (a 20% supply cut, a burden of about 3% of GDP) the effect is marginal: one additional importer enters crisis. Under a severe shock (a burden of about 15% of GDP, comparable to the worst historical oil episodes) the mean importer debt-crisis duration rises about 2.7-fold and four further countries enter sovereign crisis (Fig. 7). The response is threshold-stepped, and the energy-system model produces no sovereign crises at any magnitude. This is the mechanism behind episodes such as Sri Lanka's 2022 default.

B.3 Crop yield shock (anchor: AgMIP and AR6)

Crop assessments in the AgMIP project and the IPCC Sixth Assessment stop at yield decline and a food-security index. GIM carries the same yield loss further, into domestic instability: cutting the production and reserves of the most income-vulnerable importers opens a food-supply gap that raises protest pressure, and the escalation accelerates as the loss deepens. This is the nonlinear "crossing into high impact" documented for food prices and the Arab Spring [12] (a threshold FAO food-price index of 210). At a severe 50% regional yield loss, food-affordability stress rises by about 0.09 and protest pressure by about 0.014 on the vulnerable subset (Fig. 8); further along the chain the response attenuates toward regime fragility, and the endpoint is domestic instability rather than interstate conflict.

Across the three channels (Fig. 9) GIM exhibits a positive, measurable slope on an axis where every sectoral model is, by construction, flat at zero. That gap — not any single number — is the integration dividend, and it is exactly the dimension the coverage table summarizes qualitatively.

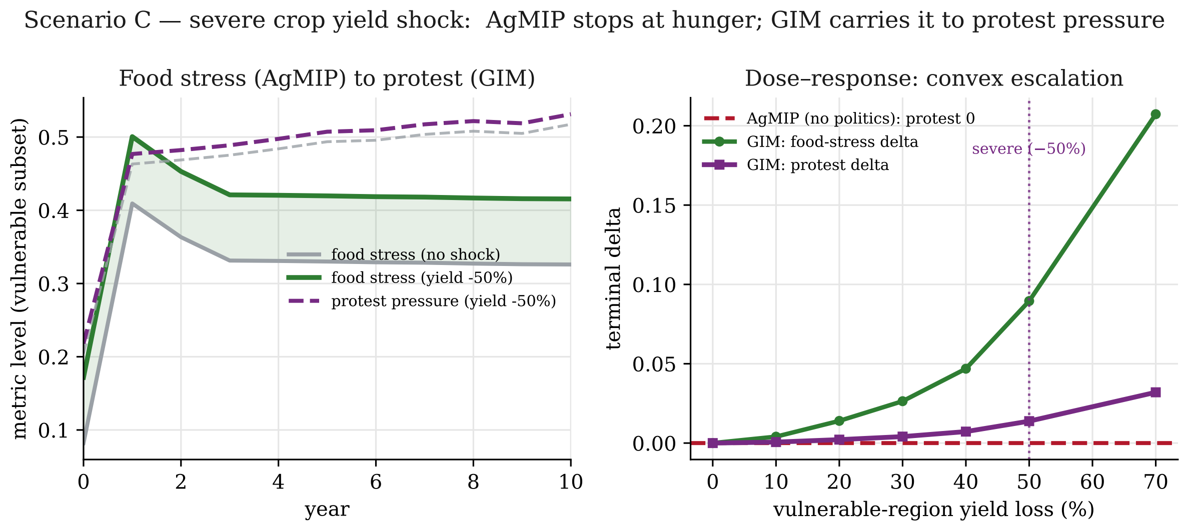

Figure 8: Crop-yield-shock scenario. Left: food-affordability stress and protest pressure on the vulnerable subset. Right: the response to the depth of the yield loss is convex against AgMIP's zero response on the protest axis.

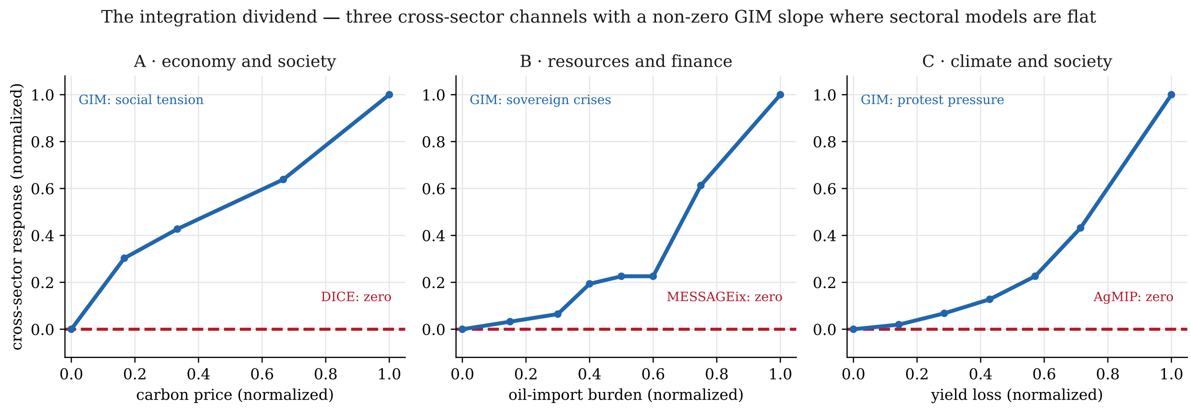

Figure 9: The integration dividend. Each panel is one cross-sector channel, normalized to its own maximum; the dashed line is every sectoral model's zero response on that axis. GIM's slope is smooth for the economy and society channel, threshold-stepped for the resources and finance channel, and convex for the climate and society channel.

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